No. of Functional Features Included in the Solution

Date 05 October 2026

Team ID UNM-293

Project ID 2026-2430

Project Name FitBuddy-AI fitness plan Generator using Gemini Models

Name Kaviya sree.T

Maximum Marks 5 Marks

Functional Features Overview:

This document lists the functional features included in the FitBuddy project solution. Each feature is described with its purpose, implementation status, module/component, marks, and contribution to the overall AI-based fitness-plan generation solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks	Contribution
1	User Fitness Goal Input	Allows users to enter their fitness goal and basic requirements.	User Input Module	Done	1	Provides the information required to generate a personalized plan.
2	Fitness Preference Collection	Collects preferences such as workout type, schedule and available equipment.	User Input Module	Done	1	Improves personalization of the generated fitness plan.
3	Prompt Generation	Converts user requirements into a structured prompt for Gemini Models.	Prompt Builder	Done	1	Connects user requirements with the AI generation process.
4	Gemini AI Integration	Sends the structured prompt to Gemini Models and receives the AI response.	Gemini API Service	Done	1	Provides the core AI capability of FitBuddy.
5	Personalized Fitness Plan	Generates a structured fitness plan based on the user's submitted requirements.	Plan Generator	Done	1	Delivers the main output and solves the project's core problem.
6	Plan Formatting	Presents the generated fitness plan in a clear and easy-to-follow format.	Plan Formatter	Done	1	Makes the AI-generated plan practical and readable.
7	Input Validation & Error Handling	Validates required inputs and handles API or generation failures.	Validation / Error Module	In Progress	1	Improves reliability and user experience.
8	Testing & Verification	Tests different fitness requirements and verifies generated plan output.	Testing Module	In Progress	1	Ensures the main FitBuddy workflow works as expected.

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	6
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	Initial testing in progress

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	**User Interface (UI)**	2	Fitness goal input; preference and requirement collection; plan display.
2	**Backend / Logic**	3	Input validation; prompt generation; fitness-plan formatting.
3	**Database / Storage**	0 / Not required for core prototype	Persistent storage can be added if required.
4	**API / Integration**	1	Gemini Models / Gemini API integration.
5	**Security / Authentication**	0 / Not specified in current scope	Authentication is not included in the supplied project scope.

Assigned Work – Kaviya sree.T

Primary responsibility: Implementing the FitBuddy AI fitness-plan generation features, including collecting and validating user fitness requirements, preparing structured prompts, integrating Gemini Models, generating and formatting personalized fitness plans, handling generation errors, testing different inputs, and documenting the completed functionality.